

Matrix Manifolds

Vector fields

Def. A (smooth) vector field ζ on a manifold M is a smooth mapping:

$$\begin{aligned}\zeta : M &\rightarrow TM \\ x &\mapsto \zeta_x\end{aligned}$$

Smoothness means:

$$\zeta_x = X_1 \partial_1 + \dots + X_n \partial_n$$

where $x \mapsto (X_1, \dots, X_n)$ is a smooth function on manifold M .

Tangent vectors can act on functions on M .

induce $\rightarrow$ vector field act on functions on M .

$$(\zeta f)(x) = \zeta_x(f). \quad \forall f \in \mathcal{F}(M), \quad x \in M$$

A vector-field is a $\mathcal{F}(M)$ -vector space (denoted $\mathcal{X}(M)$)

$$\forall \zeta, \zeta' \in \mathcal{X}(M), \quad f \in \mathcal{F}(M)$$

$$(f\zeta)_x = f(x)\zeta_x$$

$$(\zeta + \zeta')_x = \zeta_x + \zeta'_x$$

Tangent vectors as derivations.

• What is a derivation?

Example. (Derivative mapping)

$$f : \mathbb{R}^n \rightarrow \mathbb{R} \text{ function.}$$

$$\partial_i f : \mathbb{R}^n \rightarrow \mathbb{R}.$$

$$x \mapsto \partial_i f(x).$$

$(\partial_i)_x$ maps $f \in \mathcal{F}(\mathbb{R}^n)$ at x to $\partial_i f(x) \in \mathbb{R}$.

$$(\partial_i)_x : \mathcal{F}(\mathbb{R}^n) \rightarrow \mathbb{R}.$$

$$f \mapsto (\partial_i)_x f := \partial_i f(x)$$

It satisfies

$$1. \text{ } \mathbb{R}\text{-linear} : (\partial_i)_x (af + bg) = a(\partial_i)_x(f) + b(\partial_i)_x(g) \quad \forall a, b \in \mathbb{R}, \\ f, g \in \mathcal{F}(\mathbb{R}^n).$$

$$2. \text{ Leibnizian} : (\partial_i)_x (fg) = (\partial_i)_x(f) g(x) + f(x) (\partial_i)_x(g) \quad \forall f, g \in \mathcal{F}(\mathbb{R}^n).$$

Def (derivation)

A derivation at $x \in M$ is a mapping $\mathbb{Z}_x : \mathcal{F}(M) \rightarrow \mathbb{R}$ satisfying

$$1. \text{ } \mathbb{R}\text{-linear} : \mathbb{Z}_x (af + bg) = a\mathbb{Z}_x(f) + b\mathbb{Z}_x(g) \quad \forall a, b \in \mathbb{R}, \\ f, g \in \mathcal{F}(M).$$

$$2. \text{ Leibnizian} : \mathbb{Z}_x (fg) = \mathbb{Z}_x(f) g(x) + f(x) \mathbb{Z}_x(g) \quad \forall f, g \in \mathcal{F}(M)$$

With

$$(\mathbb{Z}_x + \mathbb{I}_x)(f) := \mathbb{Z}_x(f) + \mathbb{I}_x(f)$$

$$(a\mathbb{Z}_x)(f) := a\mathbb{Z}_x(f) \quad \forall f \in \mathcal{F}(M), a \in \mathbb{R}.$$

the set of all derivations at x becomes a vector space.

Prop. Tangent vectors at $x \Leftrightarrow$ derivations at x .

Proof.

(i) $\forall$ curve γ on M , $\gamma(0) = x \in M$

The mapping $\dot{\gamma}(0)$ defined by

$$\dot{\gamma}(0)f := \left. \frac{d(f(\gamma(t)))}{dt} \right|_{t=0}, \quad f \in \mathcal{F}_x(M)$$

is a derivation at x .

(ii) $\forall$ derivation $\mathbb{Z}_x$ at x .

$\exists$ curve γ on M s.t. $\gamma(0) = x \in M$, $\dot{\gamma}(0) = \mathbb{Z}_x$

for example $\gamma(t) = \varphi^{-1} \left[\varphi(0) + t \sum_i (\mathbb{Z}_x(\varphi_i) e_i) \right]$

where φ is a chart in a neighborhood of x

$$\begin{aligned} \dot{\gamma}(0)f &= \left. \frac{d(f(\gamma(t)))}{dt} \right|_{t=0} = \left. \frac{d((f \circ \varphi^{-1})(\varphi(\gamma(t))))}{dt} \right|_{t=0} \\ &= (D(f \circ \varphi^{-1})(\varphi(\gamma(t))) \cdot \left. \frac{d[\varphi(\gamma(t))]}{dt} \right|_{t=0} \end{aligned}$$

$$\begin{aligned}
&= (D(f \circ \varphi^{-1}))(\varphi(r(t))) \cdot \sum_i (\zeta_x(\varphi_i) e_i) \Big|_{t=0} \\
&= D(f \circ \varphi^{-1})(\varphi(x)) \cdot \sum_i \zeta_x(\varphi_i) e_i \\
\zeta_x f &= \zeta_x \underbrace{f \circ \varphi^{-1} \circ \varphi}_{\text{function on } \mathbb{R}^n} \\
&= D(f \circ \varphi^{-1})(\varphi(x)) \cdot \sum_i \zeta_x(\varphi_i) e_i \quad (\text{chain-rule on } \mathcal{F}(\mathbb{R}^n))
\end{aligned}$$

□

Derivation ζ_x at x is a local notion :

If $f = g$ on a neighborhood of x . Then

$$\zeta_x(f) = \zeta_x(g)$$

A global derivation on $\mathcal{F}(M)$ is a mapping $D : \mathcal{F}(M) \rightarrow \mathcal{F}(M)$ that is $\forall f, g \in \mathcal{F}(M)$

$$1. \text{ } \mathbb{R}\text{-linear} : D(af + bg) = a D(f) + b D(g) \quad , \quad a, b \in \mathbb{R}.$$

$$2. \text{ Leibnizian} : D(fg) = D(f)g + f D(g)$$

Prop. Vector fields $\iff$ global derivations.

$$\begin{array}{ccc}
\text{Explanation.} & T_x M & \longleftrightarrow \left(\begin{array}{c} \mathcal{F}(M) \rightarrow \mathbb{R} \\ f \mapsto \zeta_x f \end{array} \right) \\
\downarrow & & \\
& \left(\begin{array}{c} M \rightarrow TM \\ x \mapsto T_x M \end{array} \right) & \longleftrightarrow \left(\begin{array}{c} \mathcal{F}(M) \rightarrow \mathcal{F}(M) \\ f \mapsto \zeta(f) \end{array} \right)
\end{array}$$

$$\begin{aligned}
 D : F(M) &\longrightarrow F(M) \\
 f &\longmapsto Df \\
 &\quad x \mapsto Df(x)
 \end{aligned}$$

Alternatively, view

$$\begin{aligned}
 D : M \times F(M) &\longrightarrow \mathbb{R} \\
 (x, f) &\longmapsto D_x f
 \end{aligned}$$

vector field

$$\begin{aligned}
 \zeta : M &\longrightarrow TM \\
 x &\longmapsto \zeta_x
 \end{aligned}$$

$$D(x, \cdot) = \zeta_x \quad (\text{by vector field} \Leftrightarrow \text{derivation})$$

$$\text{Thus } D = \zeta$$

#

Differential

Up to now, we can only compute differential of a mapping defined on an Euclidean space.

What if the mapping is defined on a manifold M ?

$F : M \rightarrow N$ smooth mapping between two manifolds

ζ_x : tangent vector at $x \in M$

Define the mapping $DF(x)$ as

$$(DF(x)[\zeta])f := \zeta(f \circ F), \quad f \in \mathcal{F}(N)$$

it is a tangent vector to N at $F(x)$.

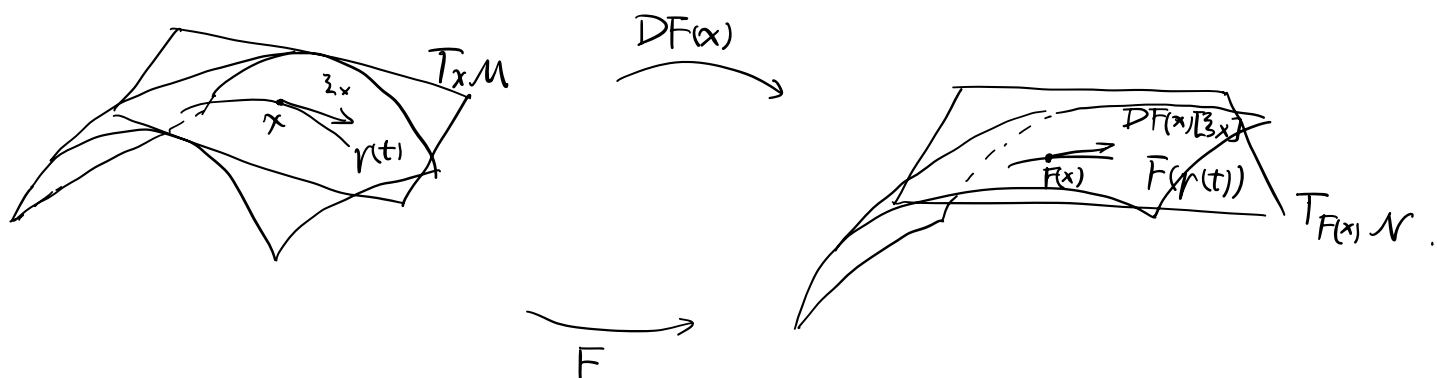
(If ζ_x is realized by some curve γ in M .

then $DF(x)[\zeta]$ is realized by the curve $F \circ \gamma$ in N .)

This mapping

$$\begin{aligned} DF(x) : T_x M &\longrightarrow T_{F(x)} N \\ \zeta &\longmapsto DF(x)[\zeta] \end{aligned}$$

is a linear mapping, called the differential (differential map, derivative, or tangent map) of F at x .



Specifically, if $\mathcal{N}$ is a vector space $\mathcal{E}$, then the canonical identification $T_{F(x)} \mathcal{E} \simeq \mathcal{E}$ yields

$$DF(x)[z_x] = \sum_i (z_x F^i) e_i$$

$\downarrow$ basis of $\mathcal{E}$

Detail: $T_{F(x)} \mathcal{E} \simeq \mathcal{E} \Rightarrow DF(x)[z_x] \in \mathcal{E}$.

$$DF(x)[z_x] f = z_x (f \circ F)$$

If $f(y) = y^i$ is the linear coordinate function

(y^i is the i -th component of vector y)

then

$$DF(x)[z_x] f = z_x F^i$$

Since $DF(x)[z_x] \in T_{F(x)} \mathcal{E} \simeq \mathcal{E}$

$$DF(x)[z_x] f = \sum_i (z_x F^i) e_i$$

Differential of a function

If $\mathcal{N} = \mathbb{R}$, then $F \in \mathcal{F}_x(\mathcal{M})$, we simply have

$$DF(x)[\zeta_x] = \zeta_x F$$

using the identification $T_x \mathbb{R} \cong \mathbb{R}$.

We will often use $DF(x)[\zeta_x]$ as an alternative notation for $\zeta_x F$. (This simply means the directional derivative of F along ζ_x)

Function between vector spaces

If both $\mathcal{M}$ and $\mathcal{N}$ are linear manifolds

$$T_x \mathcal{M} \cong \mathcal{M}, \quad T_y \mathcal{N} \cong \mathcal{N}.$$

$$\begin{aligned} DF(x)[\zeta_x] &= \sum_i (\zeta_x F^i) e_i && (T_y \mathcal{N} \cong \mathcal{N}). \\ &= \sum_i \lim_{t \rightarrow 0} \frac{F^i(x + tv) - F^i(x)}{t} e_i && \left(\begin{array}{l} T_x \mathcal{M} \cong \mathcal{M} \\ \zeta_x = v \end{array} \right) \\ &= \lim_{t \rightarrow 0} \frac{F(x + tv) - F(x)}{t} \end{aligned}$$

Extend from tangent vectors to vector fields.

We have discussed the differential of a mapping on a fixed $x \in M$.

Now, we allow x vary in M .

The tangent vector z_x at x becomes a vector field z on M .

$$\begin{aligned} DF[z] : M &\longrightarrow T\mathcal{N} \\ x &\longmapsto DF(x)[z_x] \end{aligned}$$

In particular, given a real-valued function f on M and a vector field z on M .

$$Df[z] = z f$$

Tangent vectors to embedded submanifolds

M : embedded manifold of vector space E .

γ : curve in M , $\gamma(0) = x$.

Define

$$\gamma'(0) := \lim_{t \rightarrow 0} \frac{\gamma(t) - \gamma(0)}{t}$$

well-defined since $\gamma(t) \in E$. (Strictly, write $i(\gamma(t)) - i(\gamma(0))$)

Then $\gamma'(0) \in T_x E \simeq E$

Since γ is a curve in M , it also induces a tangent vector

$$\dot{\gamma}(0) \in T_x M$$

If $\bar{f} \in U \rightarrow \mathbb{R}$ (U : neighborhood of x in E)

$$f := \bar{f}|_{U \cap M}$$

then

$$\dot{\gamma}(0) f = \left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} = \left. \frac{d}{dt} \bar{f}(\gamma(t)) \right|_{t=0} = D\bar{f}(x)[\gamma'(0)]$$

This yields a natural identification

$$T_x M \longleftrightarrow \{ \gamma'(0) : \gamma \text{ curve in } M, \gamma(0) = x \} \subset T_x E = E$$

(Now, we can write $\dot{\gamma}(t)$, $\gamma'(t)$ and $\frac{d}{dt} \gamma(t)$ interchangeably in the context of embedded manifolds. We use "=" for this identification.)

When M is defined as a level set of a constant-rank function $F: \Sigma \rightarrow \mathbb{R}^n$, i.e. $M = F^{-1}(c)$, we have

$$T_x M = \ker(DF(x))$$

Proof.

" $\subseteq$ ": γ : curve in M with $\gamma(0) = x$, we have $F(\gamma(t)) = c, \forall t$.

Hence

$$DF(x)[\dot{\gamma}(0)] = \left. \frac{d(F(\gamma(t)))}{dt} \right|_{t=0} = 0$$

which shows $\dot{\gamma}(0) \in \ker(DF(x))$

" $=$ ": Suppose F has constant-rank $k < d := \dim(\Sigma)$,

by Prop 3.3.4 (subimmersion thm), $F^{-1}(c)$ is a closed embedded submanifold of dim $d - k$. $\Rightarrow \dim T_x M = d - k$

$$\dim \ker(DF(x)) = d - \text{rank}(DF(x)) = d - k.$$

□

Example 1. (Tangent space to a sphere)

Method 1: $t \mapsto x(t)$: curve in S^{n-1} , $x(0) = x_0 \in S^{n-1}$.

Since $x(t) \in S^{n-1}$, $\forall t$,

$$x^T(t) x(t) = 1 \quad \forall t.$$

Differentiating w.r.t. t yields

$$\dot{x}^T(t) x(t) + x^T(t) \dot{x}(t) = 0,$$

hence $\dot{x}(0)$ is an element of

$$\{ z \in \mathbb{R}^n : x_0^T z = 0 \} =: V$$

This shows $T_{x_0} S^{n-1} \subseteq V$.

Conversely, let $z \in V$, then the curve

$$t \mapsto x(t) := (x_0 + tz) / \|x_0 + tz\|$$

is on S^{n-1} with $\dot{x}(0) = z$. Hence $V \subseteq T_{x_0} S^{n-1}$.

In conclusion, $T_x S^{n-1} = \{ z \in \mathbb{R}^n : x^T z = 0 \}$

Method 2:

Consider the function $F: \mathbb{R}^n \rightarrow \mathbb{R}$, $x \mapsto x^T x - 1$.

$S^{n-1} = \{ x \in \mathbb{R}^n : F(x) = 0 \}$ and F is full rank on S^{n-1} (rank 1).

Thus,

$$\begin{aligned} T_x S^{n-1} &= \ker(DF(x)) = \{ z \in \mathbb{R}^n : x^T z + z^T x = 0 \} \\ &= \{ z \in \mathbb{R}^n : x^T z = 0 \} \end{aligned}$$

Example 2. (Orthogonal Stiefel manifold)

Consider the orthogonal Stiefel manifold

$$\text{St}(p, n) = \{ X \in \mathbb{R}^{n \times p} : X^T X = I_p \}$$

as an embedded submanifold of $\mathbb{R}^{n \times p}$.

Method 1:

$$X_0 \in \text{St}(p, n).$$

$$t \mapsto X(t) : \text{curve in } \text{St}(p, n) \quad , \quad X(0) = X_0$$

$$X^T(t) X(t) = I_p \quad , \quad \forall t$$

$$\Rightarrow \dot{X}^T(t) X(t) + X^T(t) \dot{X}(t) = 0$$

$$\Rightarrow \dot{X}(0) \text{ belongs to}$$

$$\{ Z \in \mathbb{R}^{n \times p} : X_0^T Z + Z^T X_0 = 0 \} =: V$$

$$\Rightarrow T_{X_0} \text{St}(p, n) \subseteq V$$

Conversely, $\forall Z \in V$,

$$X(t) = (X_0 + tZ) \left[(X_0 + tZ)^T (X_0 + tZ) \right]^{-\frac{1}{2}}$$

is a curve on $\text{St}(p, n)$ with

$$\dot{X}(t) = Z$$

$$\Rightarrow V \subseteq T_{X_0} \text{St}(p, n)$$

$$\text{Thus } T_X \text{St}(p, n) = \{ Z \in \mathbb{R}^{n \times p} : X^T Z + Z^T X = 0 \}$$

$$\left(\begin{aligned} \dim &= np - \frac{p(p+1)}{2} \\ &= (n-p)p + \frac{p(p-1)}{2} \end{aligned} \right)$$

Method 2 :

$$F : X \mapsto X^T X \quad \left(\text{constant rank } \frac{p(p+1)}{2} \right)$$

$$St(p, n) = F^{-1}(I_p)$$

$$\begin{aligned} T_X St(p, n) &= \text{Ker}(DF(X)) \\ &= \{ Z \in \mathbb{R}^{n \times p} : X^T Z + Z^T X = 0 \} \end{aligned}$$

Method 3 :

Notice $X(t)$ has full column rank.

$X_\perp(t) : n \times (n-p)$ matrix s.t. $\text{span}(X_\perp(t))$ is the orthogonal complement of $\text{span}(X(t))$.

Then $\dot{X}(t) \in \mathbb{R}^{n \times p}$ can be represented as

$$\dot{X}(t) = X(t)\Omega(t) + X_\perp(t)K(t), \quad \begin{aligned} \Omega(t) &\in \mathbb{R}^{p \times p} \\ K(t) &\in \mathbb{R}^{(n-p) \times p} \end{aligned}$$

Substitute into

$$\dot{X}^T(t) X(t) + X^T(t) \dot{X}(t) = 0$$

yields

$$\Omega(t)^T + \Omega(t) = 0$$

counting dimensions, we deduce

$$T_X St(p, n) = \{ X\Omega + X_\perp K : \Omega^T = -\Omega, K \in \mathbb{R}^{(n-p) \times p} \}$$

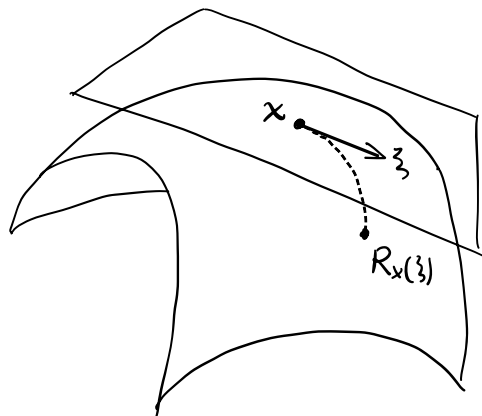
Example 3. (Orthogonal group)

As a special case, $O_n = St(n, n)$,

$$T_O O_n = \{ Z = U\Omega : \Omega^T = -\Omega \} = \cup S_{\text{skew}}(n)$$

Retractions

Generalizes the notion of moving in the direction of a tangent vector, while staying on the manifold.



A retraction R at x (denoted by R_x), is a mapping from $T_x M$ to M with a local rigidity condition that preserves the gradient at x .

Def (retraction)

A retraction on a manifold M is a smooth mapping R

$$R : TM \rightarrow M$$

with the following properties.

Let R_x denote the restriction of R to $T_x M$ ($R_x : T_x M \rightarrow M$)

(i) $R_x(0_x) = x$, where 0_x denote the zero element of $T_x M$.

(ii) With the canonical identification $T_{0_x} T_x M \cong T_x M$,

R_x satisfies

$$DR_x(0_x) = \text{id}_{T_x M}. \quad (\text{local rigidity condition})$$

Explanation on (ii) :

$$R_x : T_x M \rightarrow M, \quad R_x(0_x) = x$$

$$DR_x(0_x) : T_{0_x} T_x M \rightarrow T_x M$$

$T_x M$ is a vector space. $\Rightarrow$ Natural identification $T_{0_x} T_x M \cong T_x M$.

Another point of view on (ii) :

For every tangent vector $\zeta \in T_x M$, the retraction gives a curve on M :

$$\gamma_\zeta : t \mapsto R_x(t\zeta), \quad \gamma_\zeta(0) = x$$

This curve gives a tangent vector $\dot{\gamma}_\zeta(0)$ at x .

$$\begin{aligned} \dot{\gamma}_\zeta(0) f &= \left. \frac{d}{dt} (f \circ \gamma_\zeta) \right|_{t=0} \\ &= \left. \frac{d}{dt} (f \circ R_x(t\zeta)) \right|_{t=0} \quad \swarrow \text{viewed as vector space} \\ &= \zeta(f \circ R_x) \quad \xleftarrow{\text{viewed as derivation}} (\zeta \in T_x M) \\ &= DR_x(0_x)[\zeta] f \quad (\text{use local rigidity cond.}) \\ &= \text{id}_{T_x M}[\zeta] f \\ &= \zeta f \end{aligned}$$

$$\Leftrightarrow \dot{r}_3(0) = \zeta$$

Thus the local rigidity condition is saying that the curve r_3 induced by the retraction gives tangent vector ζ .

Moving along this curve r_3 is thought of as moving in the direction ζ while constrained to the manifold M .

Remark.

In optimization, retracts can transform cost functions defined in a neighborhood of $x \in M$ into cost functions defined on the vector space $T_x M$.

Given $f : M \rightarrow \mathbb{R}$ and retraction $R : TM \rightarrow M$
 $R_x : T_x M \rightarrow M$.

The pullback of f through R is

$$\hat{f}_x := R_x^* f := f \circ R_x$$

Now, $\hat{f}_x$ is a real-valued function on vector space $T_x M$.

~~Because of the local rigidity condition $DR_x(0_x) = \text{id}_{T_x M}$, and the canonical identification $T_{0_x} T_x M \simeq T_x M$~~

$$D\hat{f}_x(0_x) = Df(x)$$

Because

$$\forall g \in \mathcal{F}(\mathbb{R}), \quad \xi \in T_{0_x} T_x M$$

$$\begin{aligned} (D\hat{f}_x(0_x)[\xi])g &\equiv \xi(g \circ \hat{f}_x)_{(0_x)} \\ &= \xi(g \circ f \circ R_x)_{(0_x)} \\ &= \xi(g \circ f)(R_x(0_x)) \\ &= \xi(g \circ f)(x) \\ &= Df(x)[\xi](g) \end{aligned}$$

$$\Rightarrow D\hat{f}_x(0_x) = Df(x)$$

Retractions on embedded submanifolds.

$\mathcal{M}$: embedded submanifold of a vector space $\mathcal{E}$.

$T_x \mathcal{M}$: linear subspace of $T_x \mathcal{E} \simeq \mathcal{E}$

Notation : $x + \zeta \in \mathcal{E}$, where

1) $x \in \mathcal{M} \subset \mathcal{E}$

2) $\zeta \in T_x \mathcal{M} \subset T_x \mathcal{E} \simeq \mathcal{E}$

Idea to define retraction :

Step 1. moving along ζ to get the point $x + \zeta$ in the linear embedding space $\mathcal{E}$.

Step 2. "projecting" the point $x + \zeta$ back to the manifold $\mathcal{M}$.

In the context of matrix manifolds , "step 2" can be based on matrix decompositions.

General theory of decomposition-based retractions.

$\mathcal{M}$: embedded manifold of a vector space $\mathcal{E}$.

$\mathcal{N}$: an abstract manifold s.t. $\dim(\mathcal{M}) + \dim(\mathcal{N}) = \dim(\mathcal{E})$

Assume there is a diffeomorphism

$$\begin{aligned} \phi : \mathcal{M} \times \mathcal{N} &\rightarrow \mathcal{E}_* & (\mathcal{E}_* : \text{open subset of } \mathcal{E}) \\ (F, G) &\mapsto \phi(F, G) \end{aligned}$$

with a neutral element $I \in \mathcal{N}$ s.t

$$\phi(F, I) = F, \quad \forall F \in \mathcal{M}$$

Prop. Under the above assumptions on ϕ , the mapping

$$R_X(\xi) := \pi_1(\phi^{-1}(X + \xi))$$

where $\pi_1 : \mathcal{M} \times \mathcal{N} \rightarrow \mathcal{M}$

$$(F, G) \mapsto F$$

is the projection onto the first component,

defines a retraction on $\mathcal{M}$.

Proof. $\mathcal{E}_*$ open $\Rightarrow X + \xi \in \mathcal{E}_*$ for all ξ in some neighborhood of 0_X .

ϕ^{-1} defined on the whole $\mathcal{E}_*$

$\Rightarrow R_X(\zeta)$ defined $\forall \zeta$ in a neighborhood of the origin $T_X M$.

R is smooth

$$R_X(0_X) = X \quad \text{direct}$$

Local rigidity : $\forall \zeta \in T_X M$.

$$D_1 \phi(X, I)[\zeta] = D\phi(X, I)[(\zeta, 0)] = \zeta$$

Since $\pi_* \circ \phi^{-1}(\phi(F, I)) = F$, $\forall F \in M$, it follows that

if we take the derivative at $X \in M$ in the direction ζ :

$\forall \zeta \in T_X M$.

$$D \text{id}_M(X)[\zeta] = D(\pi_* \circ \phi^{-1})(\phi(X, I))[D_1 \phi(X, I)[\zeta]]$$

$$\text{Use } \phi(X, I) = X, \quad D_1 \phi(X, I)[\zeta] = \zeta$$

$$\Rightarrow \zeta = D(\pi_* \circ \phi^{-1})(X)[\zeta] = DR_X(0_X)[\zeta]$$

which proves that R_X is a contraction.

□

Example 1 (Retraction on the sphere S^{n-1})

Let $\mathcal{M} = S^{n-1}$, $\mathcal{N} = \{x \in \mathbb{R} : x > 0\}$, consider

$$\phi : \mathcal{M} \times \mathcal{N} \rightarrow \mathbb{R}_*^n : (x, r) \mapsto xr$$

It is straightforward to verify that ϕ is a diffeomorphism.

$$R_x(\zeta) = \frac{x + \zeta}{\|x + \zeta\|} \quad \forall \zeta \in T_x S^{n-1}$$

In this case, $R_x(\zeta)$ is the point of S^{n-1} that minimizes the distance to $x + \zeta$. #

Notation

$\mathbb{R}_*^{n \times p}$: the set of all $n \times p$ matrices whose columns are linearly independent

$\mathbb{R}_*^{n \times p}$ is an open subset of $\mathbb{R}^{n \times p}$ since its complement

$$\{X \in \mathbb{R}^{n \times p} : \det(X^T X) = 0\}$$

is closed.

Example. (Retraction on the orthogonal group)

Let $\mathcal{M} = O_n$ be the orthogonal group.

$$\mathcal{E} = \mathbb{R}^{n \times n}, \quad \mathcal{E}_* = \mathbb{R}_*^{n \times n}$$

Method 1 : QR decomposition

QR : For $A \in \mathbb{R}_*^{n \times n}$,

$$A = QR$$

$Q \in O_n$, $R \in \text{Supp}_+(n)$ (upper triangular with strictly positive diagonal elements)

Define ϕ as the inverse of the QR decomposition :

$$\phi : O_n \times \text{Supp}_+(n) \rightarrow \mathbb{R}_*^{n \times n}$$

$$(Q, R) \mapsto QR$$

Let $qf := \pi_1 \circ \phi^{-1}$ sends a matrix to the Q factor of its QR decomposition

Check that ϕ satisfies all assumptions of Prop. :

- Neutral element : identity matrix.

$$\phi(Q, I) = Q \quad \forall Q \in O_n$$

- ϕ is bijective : existence and uniqueness of QR.
- ϕ is smooth : It is the restriction of a smooth map (matrix product) to a submanifold.
- ϕ^{-1} is smooth : a) Q is obtained by a Gram-Schmidt process, it is C^∞ on $\mathbb{R}_*^{n \times n}$.

(It involves :

i) normalization : $v_i / \|v_i\|$

ii) projection : $v_i - \sum_{j < i} \langle v_i, u_j \rangle u_j$

where v_i are columns of A ,

u_j are orthonormalized columns.

Since $R_*^{n \times n}$ is full-rank, this ensures

i) No division by 0.

ii) All projections are well-defined

Each step is a smooth operation on entries of A ,

Thus the Gram-Schmidt process depends smoothly on A

b) The second component $R = Q^{-1}A$

Thus all assumptions hold.

Since

$$T_x O_n = \{ Z = X\Omega : \Omega^T = -\Omega \} = \cup S_{\text{skew}(n)}$$

Consequently,

$$\begin{aligned} R_x(X\Omega) &:= \text{qf}(X + X\Omega) = \text{qf}(X(I + \Omega)) \\ &= X \text{qf}(I + \Omega) \end{aligned}$$

is a retraction on the orthogonal group O_n .

Method 2: (Polar decomposition)

$$A = QP$$

$Q \in O_n$, $P \in S_{\text{sym}+}(n)$: all symmetric positive definite matrices of size n

The inverse of polar decomposition is

$$\begin{aligned}\phi : O_n \times S_{\text{sym}+}(n) &\rightarrow \mathbb{R}_{*}^{n \times n} \\ (Q, P) &\mapsto QP\end{aligned}$$

$$\text{We have } \phi^{-1}(A) = (A(A^T A)^{-\frac{1}{2}}, (A^T A)^{\frac{1}{2}})$$

This shows that ϕ is a diffeomorphism, thus

$$\begin{aligned}R_X(X\Omega) &:= X(I+\Omega) \left[(X(I+\Omega))^T (X(I+\Omega)) \right]^{-\frac{1}{2}} \\ &= X(I+\Omega) (I-\Omega^2)^{-\frac{1}{2}}\end{aligned}$$

$$\left\{ \begin{array}{l} \text{Note: } (X(I+\Omega))^T (X(I+\Omega)) \\ = (I+\Omega)^T X^T X (I+\Omega) \\ = (I+\Omega)^T (I+\Omega) \quad (X \text{ orthogonal}) \\ = (I-\Omega)(I+\Omega) \quad (\Omega \text{ skew-symmetric}) \end{array} \right.$$

Computing this function requires an eigenvalue decomposition of the $n \times n$ symmetric matrix $(I-\Omega^2)$.

Method 3 : Givens rotations

For $n \times n$ skew-symmetric matrix Ω

$$Giv(\Omega) = \prod_{1 \leq i < j \leq n} G(i, j, \Omega_{ij})$$

where the order of multiplication is any fixed order.

$$G(i, j, \theta) \Big|_{\{i, j\} \times \{i, j\}} = \begin{matrix} & \begin{matrix} i^{th} & j^{th} \end{matrix} \\ \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} & \begin{matrix} i^{th} \\ j^{th} \end{matrix} \end{matrix}$$

Then

$$R : T O_n \rightarrow O_n$$

$$R_x(X\Omega) = X Giv(\Omega)$$

is a retraction on O_n .

Method 4 : Cayley transform

$$R_x(X\Omega) = X \left(I - \frac{1}{2}\Omega \right)^{-1} \left(I + \frac{1}{2}\Omega \right)$$

Method 5 : Riemannian exponential mapping on O_n

$$\text{Exp}_x(X\Omega) = X \exp(\Omega)$$

Riemannian exponential mappings are always retractions.
Numerical cost for eigenvalue computation.

→ Lie-group based algorithms for approximating $\exp$ that fit the definition of a retraction.